

2016 HKDSE Physics

Paper 2 Elective Sections

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Section A Astronomy and Space Science

Question 1.1

A spacecraft is orbiting the Earth (mass M) in a circular orbit of radius r . Inside the spacecraft, a spring balance is used to measure the weight of an object (mass m). Which of the following is correct?

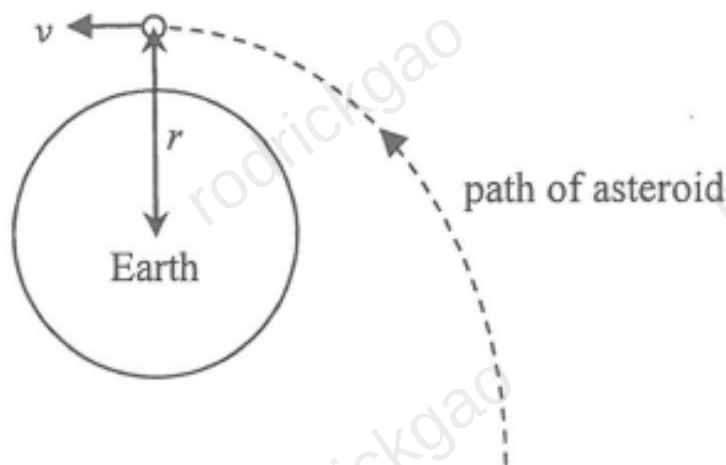
- A. Balance reading: 0; gravitational force: 0
- B. Balance reading: 0; gravitational force: GMm/r^2
- C. Balance reading: GMm/r^2 ; gravitational force: 0
- D. Balance reading: GMm/r^2 ; gravitational force: GMm/r^2

Official answer: B

The object and spacecraft are both freely falling in orbit with the same acceleration. The spring need not support the object, so its reading is zero. Gravity is not zero: GMm/r^2 provides the centripetal force.

Question 1.2

An asteroid (mass m) approaches the Earth (mass $M \gg m$) as shown. The velocity at the closest approach is v and the corresponding distance from the Earth's centre is r . Assuming no energy is lost during the asteroid's journey, what is its kinetic energy when it is very far away from the Earth?



- A. 0
- B. $\frac{1}{2}mv^2$
- C. $\frac{1}{2}mv^2 - GMm/r$
- D. $\frac{1}{2}mv^2 + GMm/r$

Official answer: C

Take gravitational potential energy as zero at infinity. Conservation of mechanical energy gives $K_{\text{infinity}} = K_{\text{close}} + U_{\text{close}} = \frac{1}{2}mv^2 - GMm/r$. The asteroid gains kinetic energy while falling into the gravitational field, so its far-away kinetic energy is smaller.

Question 1.3

Which of the following comparisons about the typical size of various celestial bodies are correct?

- (1) A star cluster is smaller than a galaxy.
- (2) A cluster of galaxies is larger than a galaxy.
- (3) A nebula is larger than a galaxy.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

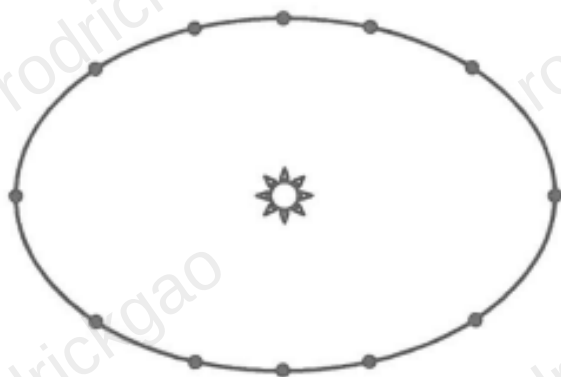
Official answer: A

A star cluster and a typical nebula are structures within a galaxy and are smaller than it. A galaxy cluster contains many galaxies and is larger. Thus (1) and (2) are correct; (3) is not.

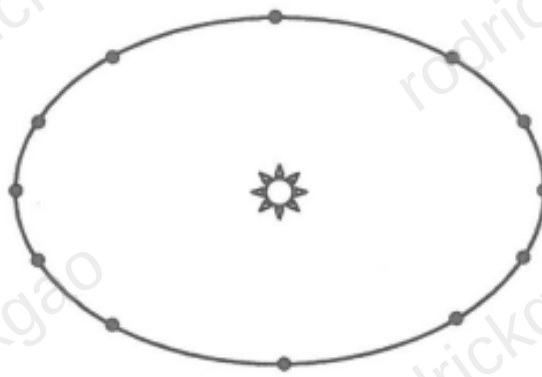
Question 1.4

Which diagram below best shows the positions of a planet orbiting a star at equal time intervals?

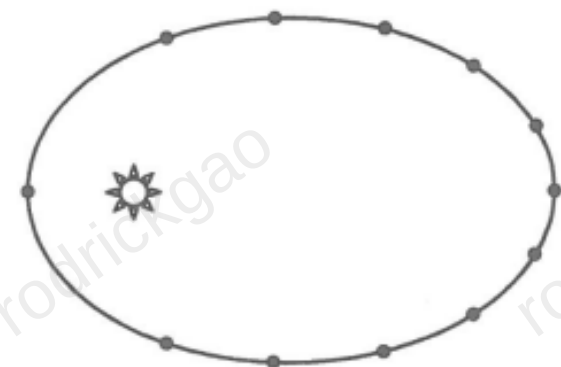
A.



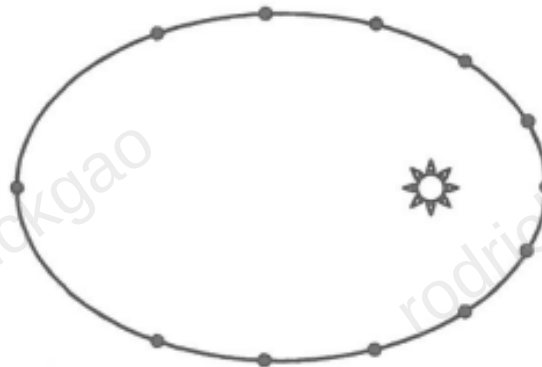
B.



C.



D.



Official answer: C

Kepler's first law puts the star at a focus, not the ellipse's centre. The second law requires equal areas in equal times: the planet is faster near the star, so the dots are farther apart there. C satisfies both conditions.

Question 1.5

A spacecraft sends a radio signal back to the Earth from 130 AU away. How long will it take for the signal to arrive at the Earth?

- A. 500 s
- B. 650 s
- C. 43333 s
- D. 65000 s

Official answer: D

Radio waves travel at c. Time = distance/speed = $(130 \times 1.50 \times 10^{11}) / (3.00 \times 10^8) = 65000$ s.

Question 1.6

The explosion of a massive star towards the end of its life results in a supernova which appears extremely bright in the sky for some time. In 1987, a supernova (SN1987A) appeared and was visible to the naked eye in the Large Magellanic Cloud which is 163000 light years away. In 1054, Chinese astronomers observed another supernova (SN1054) that appeared in the constellation of Taurus which is 6500 light years away. SN1987A happened about

- A. 933 years after SN1054.
- B. 155567 years before SN1054.
- C. 156500 years before SN1054.
- D. 162067 years before SN1054.

Official answer: B

Subtract the light-travel time from each observation date.

Emission-time difference = $(1987 - 163000) - (1054 - 6500) = -155567$ years.

Thus the explosion seen in 1987 actually occurred about 155567 years before the other explosion.

Question 1.7

Stars X and Y are of equal brightness to the naked eye. The measured parallax of star X is twice that of star Y. What is the ratio (luminosity of star X)/(luminosity of star Y)?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 2
- D. 4

Official answer: A

Distance is inversely proportional to parallax, so $d_X/d_Y = \frac{1}{2}$. Equal apparent brightness means $L_X/d_X^2 = L_Y/d_Y^2$.

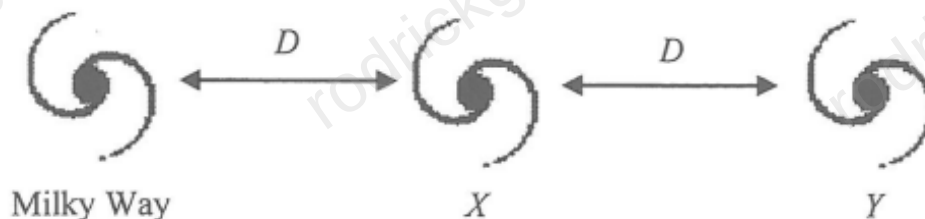
Hence $L_X/L_Y = (\frac{1}{2})^2 = \frac{1}{4}$.

Question 1.8

Three galaxies are separated by distance D as shown below. The H_α line of Galaxy X when observed from the Milky Way shows a red shift of $\Delta\lambda$.

Which of the following statements is/are correct?

- (1) The H_α line of Galaxy Y when observed from the Milky Way shows a red shift greater than $\Delta\lambda$.
- (2) The H_α line of the Milky Way when observed from Galaxy X shows no red shift.
- (3) The speed at which Galaxy X is moving away from Galaxy Y equals the speed at which Galaxy X is moving away from the Milky Way.



- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

Official answer: D

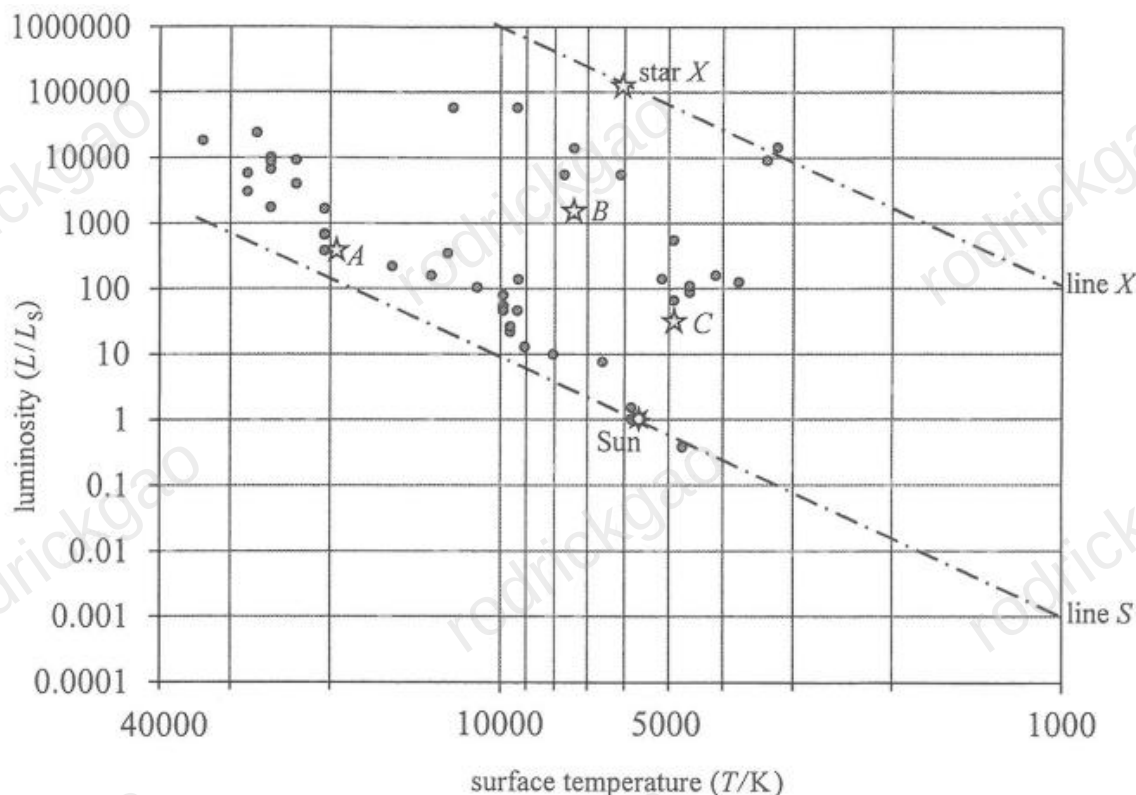
By Hubble's law, recession speed is proportional to separation. Y is twice as far from the Milky Way as X, so its red shift is greater.

Recession is mutual, so X also observes a red shift from the Milky Way. X is separated from both Y and the Milky Way by D , giving equal recession speeds. Thus (1) and (3) are correct.

Question 1 Structured Question

(a) Explain qualitatively how absolute magnitude, apparent magnitude and luminosity of a star are related. (2 marks)

The Hertzsprung-Russell diagram (HR diagram) below shows the 50 brightest stars as seen from the Earth.



(b)(i) L , R and T are the luminosity, radius and surface temperature of a star. Use Stefan's law to show that $L/L_{\odot} = (R/R_{\odot})^2 (T/T_{\odot})^4$

where $L_{\odot}$, $R_{\odot}$ and $T_{\odot}$ are the luminosity, radius and surface temperature of the Sun. State an assumption you made. (2 marks)

(b)(ii) Star X in the HR diagram has surface temperature $T = 6100$ K and luminosity $L = 126000 L_{\odot}$. Find the radius R of star X in terms of the Sun's radius $R_{\odot}$. Hence name the type of star that it belongs to.

Given: surface temperature of the Sun $T_{\odot} = 5840$ K. (3 marks)

(c)(i) Taking the logarithm of the equation in (b)(i) yields the following equation:

$$\log(L/L_{\odot}) = 4 \log T + 2 \log(R/R_{\odot}) - 4 \log T_{\odot}$$

Show that it represents a straight line in the HR diagram and all stars on the line are of the same size. The scales on both axes of the HR diagram are logarithmic and the x-axis indicates a higher temperature towards the left. $R_{\odot}$ and $T_{\odot}$ are constants.

Note: Line S and line X in the diagram are two such straight lines running from upper left to lower right containing the Sun and star X respectively. (2 marks)

(c)(ii) For stars A, B and C in the HR diagram, deduce which one is the largest. (1 mark)

Official marking scheme

Solution	Marks	Remarks
1. (a) <i>Apparent magnitude</i> is a measure of brightness and it depends on (star's) <i>luminosity</i> and distance from the Earth. When distance D is fixed (at 10 pc), it is called <i>absolute magnitude</i> which then depends only on <i>luminosity</i> .	1A 1A 2	brightness = power per unit area at the observer = luminosity / $(4\pi D^2)$
(b) (i) $L = 4\pi R^2 \sigma T^4$ $L_S = 4\pi R_S^2 \sigma T_S^4$ Assume that the Sun and the star are black bodies.	1M 1A 2	
(ii) $\frac{R}{R_S} = \left(\frac{L}{L_S}\right)^{1/2} \left(\frac{T_S}{T}\right)^2$ $\frac{R}{R_S} = (126000)^{1/2} \times \left(\frac{5840}{6100}\right)^2$ $R = 325.350364 R_S \approx 325 R_S$ Star X - (super)giant	1M 1A 1A 3	
(c) (i) $\log\left(\frac{L}{L_S}\right) = 4 \log T + 2 \log\left(\frac{R}{R_S}\right) - 4 \log T_S$ $y = \log \frac{L}{L_S} \quad x = \log T$ Accept $x = \log\left(\frac{T}{T_S}\right)$ It takes the form of a straight line $y = mx + c$ (with $m = 4$) and the y-intercept c is determined by the star radius R [Note: $c = +2 \log\left(\frac{R}{R_S}\right) - 4 \log T_S$, R_S & T_S are constants]	1A 1A 2	
(ii) B (largest)	1A 1	

(a) Apparent magnitude measures the brightness seen from Earth, which depends on luminosity and distance through $b = L/(4\pi d^2)$. Absolute magnitude is the apparent magnitude the star would have at 10 pc, so it depends only on luminosity. A more luminous star has a smaller absolute magnitude.

(b)(i) Assume the star and Sun are spherical black bodies. Stefan's law gives $L = 4\pi R^2 \sigma T^4$ and $L_S = 4\pi R_S^2 \sigma T_S^4$. Dividing cancels $4\pi\sigma$ and gives $L/L_S = (R/R_S)^2 (T/T_S)^4$.

(b)(ii) $R/R_S = \sqrt{(L/L_S)(T_S/T)^2} = \sqrt{126000(5840/6100)^2} = 325.35$. Thus $R \approx 325 R_S$ and X is a supergiant (giant is accepted in the official scheme).

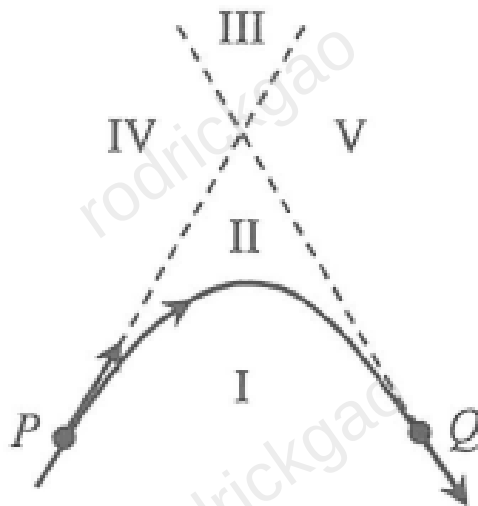
(c)(i) Let $y = \log(L/L_S)$ and $x = \log T$. Then $y = 4x + c$, where $c = 2 \log(R/R_S) - 4 \log T_S$. This is a straight line. For a fixed line, c is constant; since R_S and T_S are fixed, R is constant. It appears to slope down to the right because the temperature axis is reversed.

(c)(ii) B is largest. Compare lines parallel to the constant-radius lines: B lies on the line corresponding to the largest value of L/T^4 and hence the largest R^2 .

Section B Atomic World

Question 2.1

In the figure, the solid line is the trajectory of an α -particle scattered by a gold nucleus (not shown in figure). The dotted lines are tangents to the trajectory at points P and Q. The two dotted lines together with the trajectory divide the plane into five regions (I-V). In which region(s) can the gold nucleus be situated?



- A. I
- B. II
- C. III
- D. IV or V

Official answer: C

The positively charged α -particle is repelled by the positive gold nucleus. At both P and Q, its path bends away from the nucleus. The nucleus must lie on the opposite side of each tangent from the direction of bending. The common allowed region is III.

Question 2.2

Which statements about wave-particle duality are correct?

- (1) Interference of light is evidence that light behaves as a wave.
- (2) Photoelectric effect is evidence that light behaves as a particle.
- (3) Electron diffraction by a crystal shows that electrons behave as a wave.

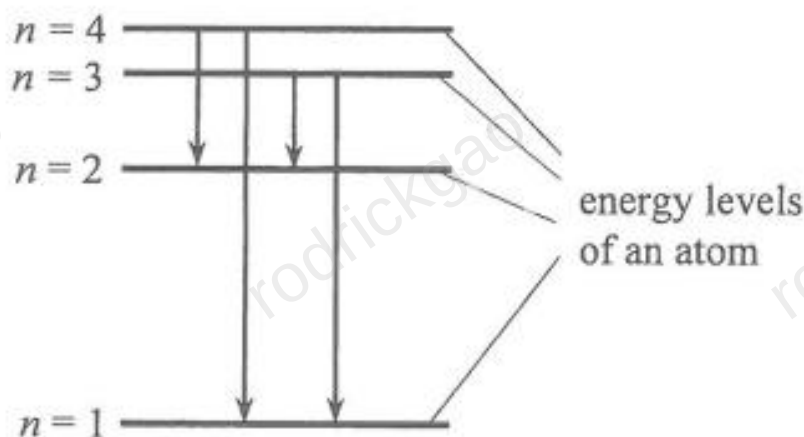
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: D

Interference demonstrates superposition and wave behaviour. The photoelectric effect shows quantized transfer of photon energy. Electron diffraction demonstrates the wave nature of matter. All three statements are correct.

Question 2.3

The figure shows four energy levels of an atom drawn approximately to scale. Which emission spectrum below best corresponds to the four electron transitions indicated? The spectra are arranged with increasing frequency to the right.



A.



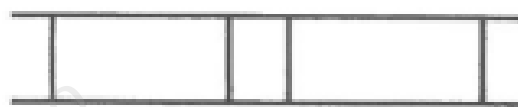
B.



C.



D.

**Official answer: C**

Each emitted photon has hf equal to the energy-level drop. The two transitions ending at $n = 2$ have smaller energies and form a low-frequency pair. The two ending at $n = 1$ have larger energies and form a high-frequency pair. Each pair differs by $E_4 - E_3$, so the within-pair spacings are equal: spectrum C.

Question 2.4

Which of the following electron transitions between energy levels in a hydrogen atom will emit electromagnetic radiation of the longest wavelength?

A. $n = 2$ to $n = 1$ B. $n = 3$ to $n = 2$ C. $n = 4$ to $n = 2$ D. $n = 5$ to $n = 2$ **Official answer: B**

The longest wavelength has the smallest photon energy, since $\lambda = hc/\Delta E$. With $E_n = -13.6/n^2$ eV, the $n = 3$ to $n = 2$ transition gives the smallest listed drop: $13.6(1/4 - 1/9) = 1.89$ eV.

Question 2.5

If the de Broglie waves associated with a proton and an α -particle have the same wavelength, what is the ratio of the kinetic energy of the proton to that of the α -particle?

- A. 1 : 4
- B. 4 : 1
- C. 1 : 2
- D. 2 : 1

Official answer: B

Equal de Broglie wavelength means equal momentum $p = h/\lambda$. For non-relativistic particles $K = p^2/(2m)$. The α -particle has about four times the proton's mass, so $K_{\text{proton}}/K_{\alpha} = m_{\alpha}/m_{\text{proton}} \approx 4$.

Question 2.6

The minimum angular separation between two points which can be resolved by the human eye (pupil diameter 4 mm) under normal lighting has the order of magnitude of

- A. 10^{-1} rad.
- B. 10^{-2} rad.
- C. 10^{-3} rad.
- D. 10^{-4} rad.

Official answer: D

Use the Rayleigh criterion $\theta \approx 1.22\lambda/d$. Taking visible light $\lambda \approx 5 \times 10^{-7}$ m and pupil diameter $d = 4 \times 10^{-3}$ m gives $\theta \approx 1.5 \times 10^{-4}$ rad, of order 10^{-4} rad.

Question 2.7

Transmission electron microscope (TEM) is used to observe structures of nano-scale instead of an optical microscope. This is because electron wave compared to visible light can have

- A. shorter wavelength so that its diffraction is less significant.
- B. shorter wavelength so that its diffraction is more significant.
- C. longer wavelength so that its diffraction is less significant.
- D. longer wavelength so that its diffraction is more significant.

Official answer: A

Accelerated electrons can have a de Broglie wavelength much shorter than visible light. A shorter wavelength reduces diffraction spreading and the minimum resolvable separation, improving resolution.

Question 2.8

Which of the following applications in nanotechnology utilize(s) Lotus effect?

- (1) Water-repelling fabric used in swimming suits is manufactured by nano-coating.
- (2) Glass is made self-cleaning by coating it with a water-attracting material in nanoscale.
- (3) Nano-sized zinc oxide is added to fabric as a photocatalyst for protection from dirt.

A. (1) only

B. (1) and (2) only

C. (1) and (3) only

D. (2) and (3) only

Official answer: A

The Lotus effect is water repellency associated with a hydrophobic, micro/nanostructured surface: droplets roll off and carry dirt away. This applies to (1). Water-attracting coatings and photocatalysis are different mechanisms.

Question 2 Structured Question

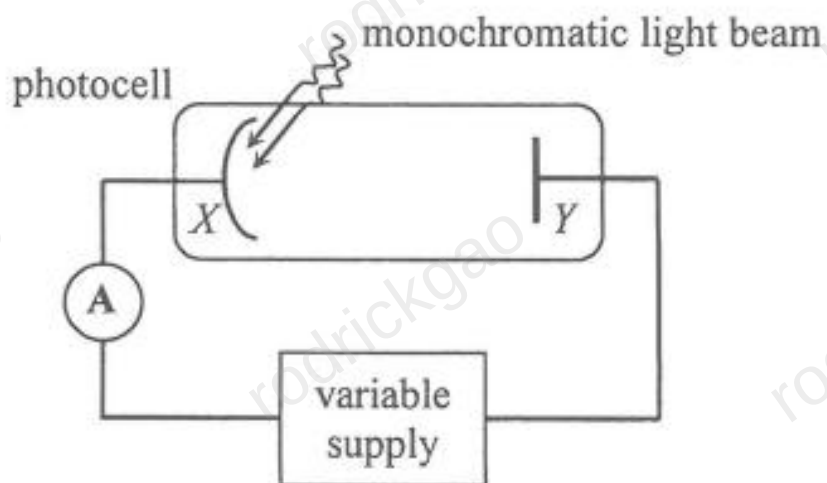
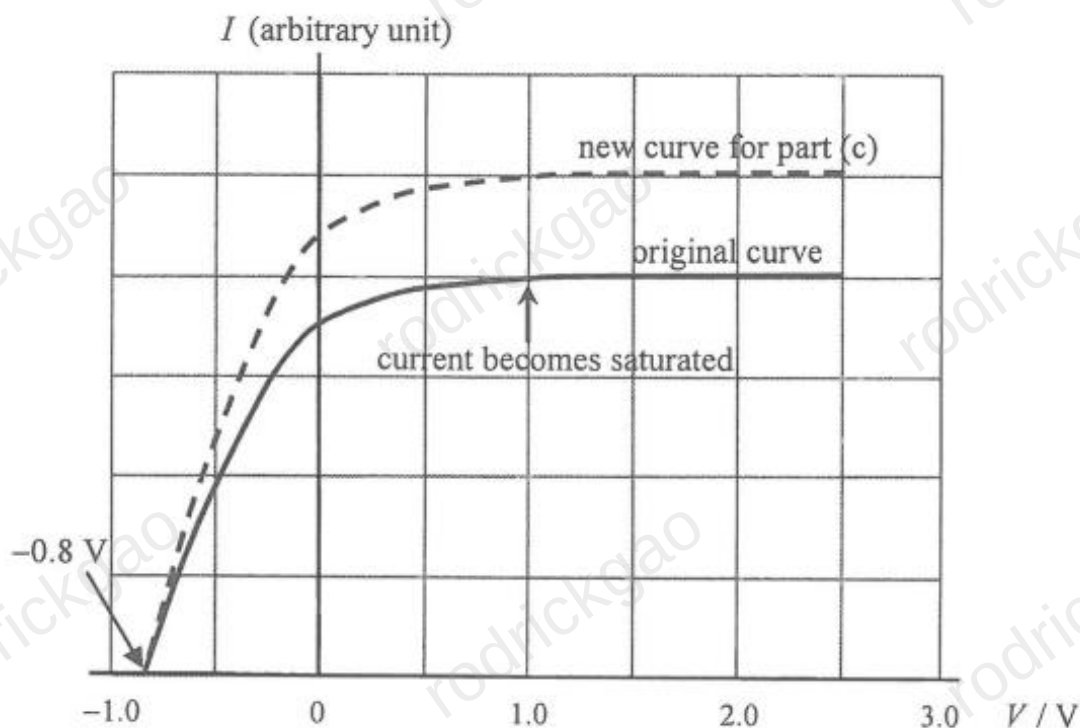


Figure 2.1

The set-up in Figure 2.1 is for the study of the photoelectric effect. A monochromatic light beam with each photon of energy 3.4 eV is directed towards the photo-sensitive cathode X of a photocell. The potential difference V across anode Y and cathode X can be changed by adjusting the variable supply. The graph shows how the photoelectric current I varies with the potential difference V .



(a)(i) The photoelectric current I becomes saturated after V reaches a certain value. Explain why this is so. (1 mark)

(a)(ii) Hence deduce the maximum kinetic energy, in eV, of the photoelectrons reaching anode Y when I is just saturated. (2 marks)

(b)(i) Find the work function, in eV, of the metal of cathode X and calculate the threshold wavelength for this metal. (3 marks)

(b)(ii) Hence explain whether yellow light of wavelength 576 nm can have photoelectric effect on cathode X. (2 marks)

(c) If the experiment is repeated with another light beam using the same photocell, a new curve (in dotted line) is obtained as shown. What can be said about this light beam's frequency and intensity? (2 marks)

Official marking scheme

Solution		Marks	Remarks
2. (a)	(i) All photoelectrons emitted (from X) can reach Y.	1A	
	Or Maximum number of photoelectrons emitted is limited by intensity of light.	1A	
	Or Limited number of photoelectrons is produced in each second.	1A	
		1	
	(ii) Maximum k.e. reaching anode $Y = (0.8 + 1.0) \text{ eV}$ $= 1.8 \text{ (eV)}$	1M 1A	
		2	
	(b) (i) $3.4 = \phi + 0.8 \Rightarrow \phi = 2.6 \text{ (eV)}$	1A	
	$\frac{hc}{\lambda} = \phi \Rightarrow \lambda = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(2.6)(1.60 \times 10^{-19})}$ $\lambda = 4.78125 \times 10^{-7} \text{ m} \approx 478 \text{ nm}$	1M 1A	
		3	
	(ii) No, as $\lambda_{\text{work}} = 576 \text{ nm} (\approx 2.16 \text{ eV}) > 478 \text{ nm} (\approx 2.6 \text{ eV})$ or threshold.	1A 1M	
		1M	
		1M	
		2	
(c)	This light beam is more intense but with the same frequency as the original one.	1A 1A	
		2	

(a)(i) At saturation all photoelectrons emitted per second are collected by the anode. Increasing the accelerating voltage cannot create more photoelectrons, so the current no longer increases.

(a)(ii) The stopping potential has magnitude 0.8 V, so the maximum initial kinetic energy is 0.8 eV. The graph shows saturation beginning at +1.0 V; an electron gains another 1.0 eV en route. Hence K_{max} at Y = 0.8 + 1.0 = 1.8 eV.

(b)(i) Einstein's equation gives $hf = \phi + K_{\text{max, initial}}$, so $\phi = 3.4 - 0.8 = 2.6 \text{ eV}$.

$$\lambda_0 = hc/\phi = (6.63 \times 10^{-34})(3.00 \times 10^8)/(2.6 \times 1.60 \times 10^{-19}) = 4.78 \times 10^{-7} \text{ m} = 478 \text{ nm}.$$

(b)(ii) No. The yellow wavelength 576 nm exceeds the threshold wavelength 478 nm, so its photon energy (about 2.16 eV) is below the 2.6 eV work function.

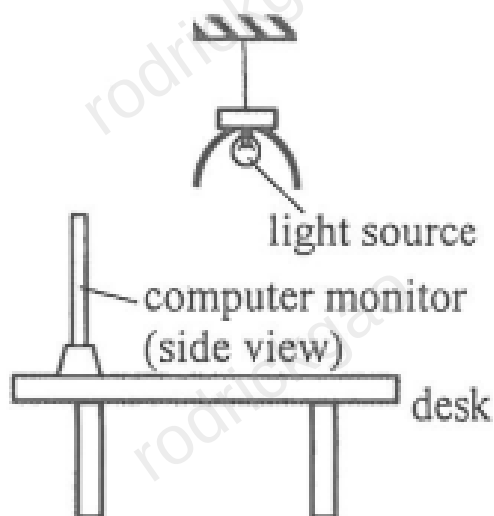
(c) Same frequency, greater intensity. The unchanged stopping potential means the same maximum photoelectron energy and hence the same photon frequency. The greater saturation current means a higher photon arrival rate at that frequency and thus greater intensity.

Section C Energy and Use of Energy

Question 3.1

In a computer room, there is a light source installed directly above each desk such that a concave reflector encloses most of the source as shown. The purpose(s) of such arrangement is/are to

- (1) maximize the amount of light reaching the desk's surface.
- (2) reduce the amount of light reaching the computer monitor so that glare can be minimized.
- (3) increase the luminous flux of the light source.



- A. (1) only
- B. (1) and (2) only
- C. (1) and (3) only
- D. (2) and (3) only

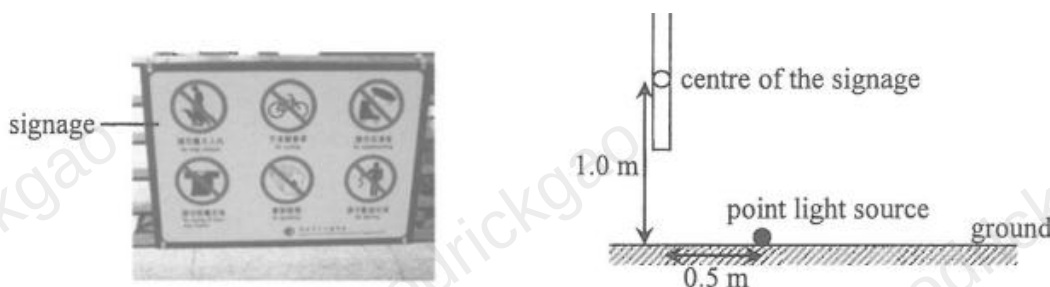
Official answer: B

The reflector redirects light onto the desk and limits light falling on the monitor, reducing glare. It does not increase the luminous flux generated by the source. Thus (1) and (2) are correct.

Question 3.2

The signage below is to be illuminated by a point light source from the ground as shown. What is the luminous flux of the source required for producing an illuminance of 200 lux at the centre of the signage?

Assume that the source emits light uniformly in all directions and neglect any reflection from the ground.



A. 7025 lm

B. 3512 lm

C. 3142 lm

D. 560 lm

Official answer: A

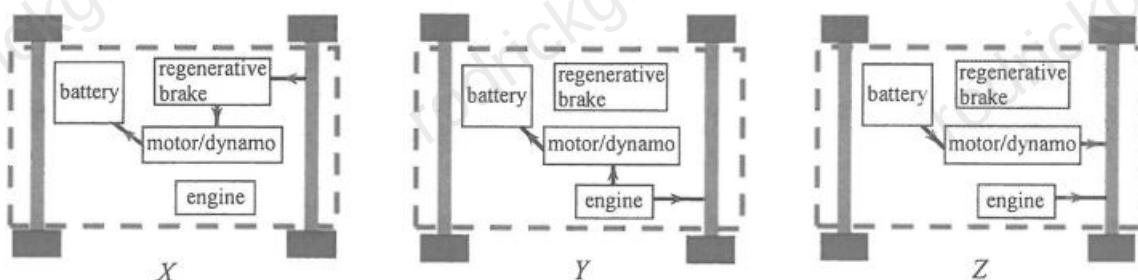
Source-to-centre distance $r = \sqrt{(1.0^2 + 0.5^2)} = \sqrt{1.25}$ m. The angle is measured to the horizontal normal of the vertical sign, so $\cos \theta = 0.5/r$.

For an isotropic point source, $E = \Phi \cos \theta / (4\pi r^2)$.

$\Phi = 4\pi(1.25)(200)/(0.5/\sqrt{1.25}) = 7025$ lm. Including the incidence-angle factor is essential.

Question 3.3

Figures X, Y and Z below show the energy flow of a hybrid car in three different situations. Which of the following correctly matches the figures with these situations?



A. Braking X; accelerating with full power Y; cruising at a constant speed Z

B. Braking Y; accelerating with full power X; cruising at a constant speed Z

C. Braking X; accelerating with full power Z; cruising at a constant speed Y

D. Braking Y; accelerating with full power Z; cruising at a constant speed X

Official answer: C

X shows regenerative braking: wheel motion drives the dynamo and energy is stored in the battery. Z shows both the engine and battery-powered motor driving the wheels for full acceleration. Y shows the engine supplying cruising power, with energy also available for charging. Therefore the order is X, Z, Y.

Question 3.4

Which statement about a solar cell is INCORRECT?

- A. When sunlight shines on a solar cell, some electrons in the semiconductor are excited to be free electrons.
- B. When a solar cell delivers power, current only flows in the interface at the p-n junction.
- C. The output voltage of a solar cell remains more or less unchanged when the incident light intensity increases.
- D. The efficiency of typical solar cells is about 10% to 20%.

Official answer: B

To deliver power, current flows around a complete circuit including the external load, not only along the p-n interface. Thus B is incorrect. The efficiency statement is retained as a statement in the 2016 examination, not a claim about all present-day solar cells.

Question 3.5

A room is cooled down by an air-conditioner and maintained at a steady temperature which is ΔT below the temperature outside. Which of the factors below would affect ΔT ?

- (1) thermal conductivity of the wall material of the room
- (2) cooling capacity of the air-conditioner
- (3) specific heat capacity of air

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: A

At steady temperature, cooling power balances heat gain. For conduction, heat gain rate = $\kappa A \Delta T / d$, so κ and cooling capacity affect the maintained ΔT . The specific heat capacity affects how quickly temperature changes, not this steady-state balance. Thus (1) and (2) are correct.

Question 3.6

Which of the following expressions take(s) the unit watt (W)?

- (1) luminous flux $\times$ area
- (2) solar constant $\times$ area
- (3) thermal transmittance $\times$ area $\times$ temperature difference

A. (2) only

B. (1) and (2) only

C. (1) and (3) only

D. (2) and (3) only

Official answer: D

(1) $\text{lm} \times \text{m}^2$ is not a unit of power.

(2) $(\text{W m}^{-2}) \times \text{m}^2 = \text{W}$.

(3) $(\text{W m}^{-2} \text{K}^{-1}) \times \text{m}^2 \times \text{K} = \text{W}$.

Therefore (2) and (3) have unit watt.

Question 3.7

An air-conditioner can remove 1 J of heat from a room to the exterior for every 0.5 J it consumes. Estimate the total thermal energy released to the exterior when the air-conditioner removes 1500 J of heat from the room.

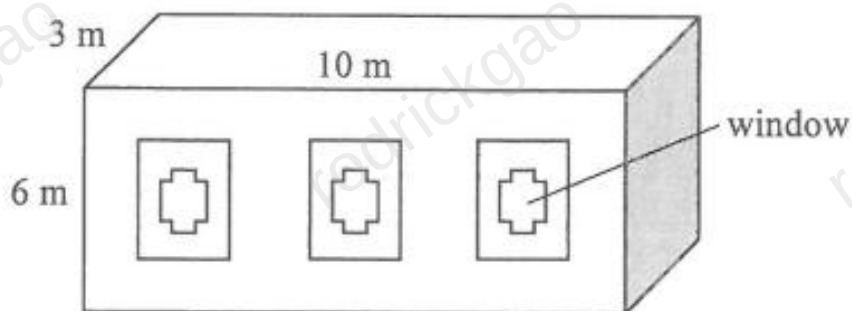
- A. 750 J
- B. 1000 J
- C. 2250 J
- D. 4500 J

Official answer: C

Electrical work input $W = 0.5 \times 1500 = 750$ J. By energy conservation, heat released outside $Q_H = Q_C + W = 1500 + 750 = 2250$ J.

Question 3.8

The Overall Thermal Transfer Value (OTTV) of the house shown is 25 W m^{-2} and the rate of heat generated due to human activities inside is 2000 W. Which cooling capacity below for an air-conditioning system is the most appropriate choice for the house?



- A. 2 kW
- B. 5 kW
- C. 10 kW
- D. 15 kW

Official answer: C

Use the exposed building envelope (four walls and roof): $A = 2(10 \times 6) + 2(3 \times 6) + 10 \times 3 = 186 \text{ m}^2$.

Total heat gain rate = OTTV $\times$ A + internal gain = $25 \times 186 + 2000 = 6650$ W. The smallest listed cooling capacity exceeding this load is 10 kW.

Question 3 Structured Question

Figure 3.1 shows a wind power station backed up by a pumped hydroelectric storage system via a control centre. Excess electrical power from the wind farm can be used to pump water from the lower reservoir to the upper one during off-peak hours. During peak hours, water runs down from the upper reservoir to drive the turbine and generator to produce electricity.

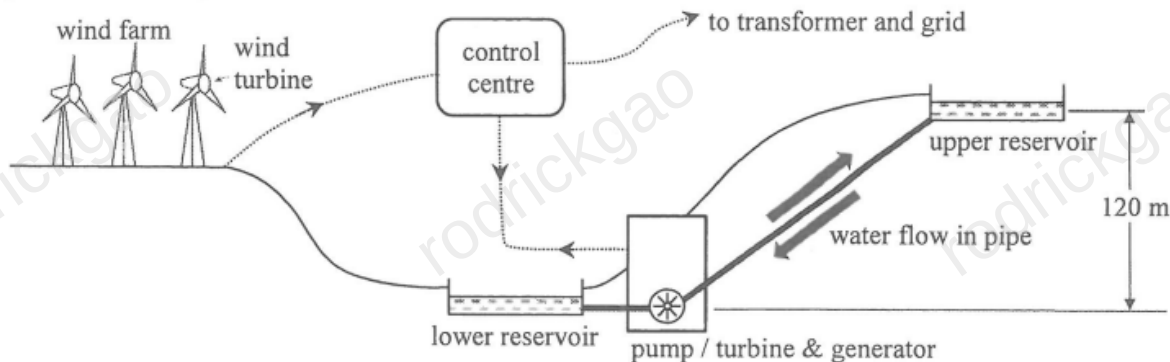
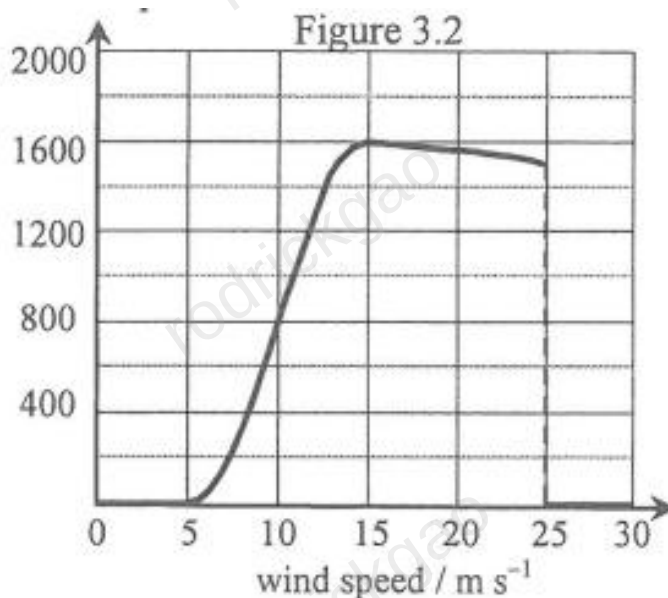


Figure 3.1

A rotor blade of each wind turbine is 30 m long. Each turbine can be automatically controlled so that the blades' rotational plane is always normal to the wind direction. The graph in Figure 3.2 shows how the electrical power output from each turbine varies with wind speed.

Electrical power output / kW



(a)(i) State the reason why practically there is no power output from the turbine when the wind speed is (I) lower than 5 m s^{-1} ; and (II) higher than 25 m s^{-1} . (2 marks)

(a)(ii) The turbine attains maximum power output when the wind speed is 15 m s^{-1} . Find the efficiency of the wind turbine in converting wind energy to electrical energy at such wind speed.

Given: density of air = 1.23 kg m^{-3} . (2 marks)

(b) There are 50 wind turbines in the wind farm for supplying electricity to a town. Figure 3.3 shows the variation of the town's electrical power demand with time of a day.

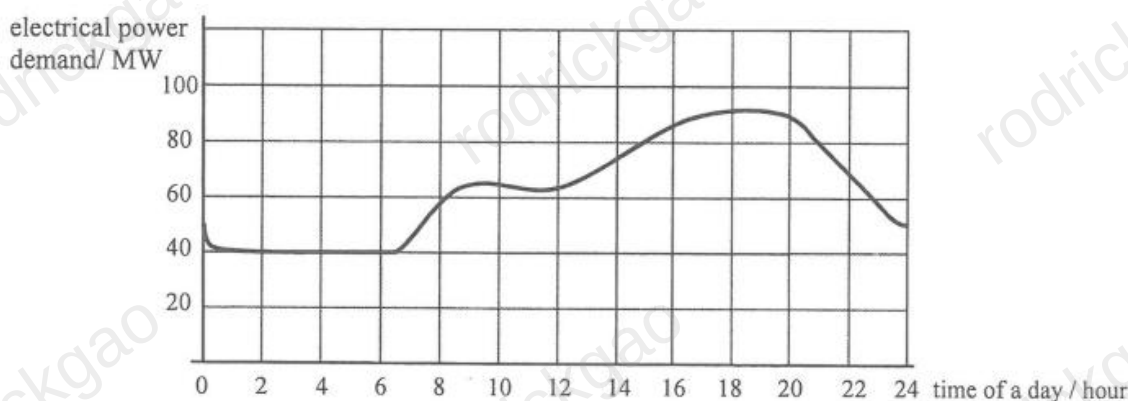


Figure 3.3

(b)(i) Using the two graphs given, find the lowest wind speed needed to meet the town's minimum demand of electrical power in a day without using the pumped hydroelectric storage system. (2 marks)

(b)(ii) Suppose that on a certain day the wind speed is always 15 m s^{-1} .

(b)(ii)(I) Estimate the total power output of the wind farm. Hence state the time period within which the pumped hydroelectric storage system needs to generate electricity for the town. (2 marks)

(b)(ii)(II) During the period of minimum demand of electrical power by the town, at what flow rate, in kg s^{-1} , is water in the lower reservoir pumped back to the upper one at a vertical height of 120 m? The overall efficiency of the pump is 80%. ($g = 9.81 \text{ m s}^{-2}$) (2 marks)

Official marking scheme

Solution	Marks	Remarks
3. (a) (i) (I) Friction between contact surfaces is too large which cannot be overcome by the wind at such speed.	1A	Note: Due to wind direction and the orientation of wind turbines, the power output of each turbine would be different in real situations.
(II) The turbine is automatically locked and stopped, otherwise the strong wind may damage the blades.	1A	
	2	
(ii) $P = \frac{1}{2} \rho A v^3 \times \eta$	1M	
$1600 \times 10^3 \text{ W} = \frac{1}{2} \times 1.23 \text{ kg m}^{-3} \times \pi (30 \text{ m})^2 \times (15 \text{ m s}^{-1})^3 \times \eta$	1A	
$\eta = 27.3 \%$	2	
(b) (i) Power required from one turbine	1M/1A	
$= \frac{40 \times 10^6}{50} = 0.8 \text{ MW or } 800 \text{ kW}$		
From the graph, wind speed needed is 10 m s^{-1} .	1A	
	2	
(ii) (I) $1600 \text{ kW} \times 50 = 80000 \text{ kW or } 80 \text{ MW}$	1M/1A	For $g = 10 \text{ m s}^{-2}$, $m = 26667 \text{ (kg s}^{-1}\text{)}$
From the graph ($>80 \text{ MW}$), 15:00 – 21:00 (i.e. 6 hours)	1A	
	2	
(II) $(80 - 40) \times 10^6 \text{ W} \times 80 \% = m \times 9.81 \text{ m s}^{-2} \times 120 \text{ m}$	1M	
$m = 2.7183 \times 10^4 \text{ (kg s}^{-1}\text{)}$	1A	
	2	

(a)(i)(I) Below 5 m s^{-1} , the wind cannot overcome the turbine's mechanical resistance/friction sufficiently to generate useful output.

(a)(i)(II) Above 25 m s^{-1} , the turbine is stopped/locked for protection against damage from strong winds.

(a)(ii) The swept area is $A = \pi r^2 = \pi(30)^2$, not the material area of the blades. Available wind power is $P_{\text{wind}} = \frac{1}{2}\rho A v^3$.

The graph gives electrical output 1600 kW. Thus $\eta = (1600 \times 10^3) / [\frac{1}{2} \times 1.23 \times \pi \times 30^2 \times 15^3] = 0.273 = 27.3\%$.

(b)(i) Minimum town demand is 40 MW, requiring $40/50 = 0.8$ MW = 800 kW per turbine. The turbine graph reaches this output at about 10 m s^{-1} .

(b)(ii)(I) Total wind output = $50 \times 1.6 \text{ MW} = 80 \text{ MW}$. The demand exceeds 80 MW from approximately 15:00 to 21:00, so hydroelectric generation is needed during that interval.

(b)(ii)(II) Excess electrical power at minimum demand = $80 - 40 = 40 \text{ MW}$. Useful pumping power = $0.80 \times 40 \times 10^6 = 32 \times 10^6 \text{ W}$.

$P_{\text{useful}} = (\Delta m / \Delta t)gh$, so $\Delta m / \Delta t = (32 \times 10^6) / (9.81 \times 120) = 2.72 \times 10^4 \text{ kg s}^{-1}$. Apply pump efficiency to the excess power, not the whole wind-farm output.

Section D Medical Physics

Question 4.1

The retina contains two types of light sensitive cells, namely, rods and cones. Which of the following statements about rods and cones is/are correct?

- (1) Rods are responsible for vision at low light levels.
- (2) Rods are less numerous than cones.
- (3) Both rods and cones are capable of colour vision.

A. (1) only

B. (1) and (2) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: A

Rods are very light-sensitive and support dim-light vision. They are more numerous than cones and do not provide colour discrimination. Cones provide colour vision. Only (1) is correct.

Question 4.2

Tom suffers from an eye defect such that his near point of accommodation is 2 m from his eyes. Spectacle lens of what power is needed to correct his near point to 0.25 m?

A. +3.5 D

B. -3.5 D

C. +4.5 D

D. -4.5 D

Official answer: A

The lens must form a virtual image at the old near point, 2 m from the eye, for an object at 0.25 m. Use $u = +0.25$ m and $v = -2.0$ m.

Power = $1/f = 1/u + 1/v = 4 - 0.5 = +3.5$ D. The positive power corresponds to a converging lens.

Question 4.3

When an aircraft takes off, some people suffer from ear pain or temporary loss of hearing. Which of the following could be the cause?

- A. The pressure on the eardrum increases drastically such that the three ear bones in middle ear cannot move.
- B. The pressure on the eardrum increases drastically such that the oval window closes.
- C. The pressure between the outer ear and middle ear is suddenly imbalanced and consequently the cochlea cannot function.

D. The pressure between the outer ear and middle ear is suddenly imbalanced and the eardrum cannot vibrate freely.

Official answer: D

Rapid change of external pressure can temporarily leave unequal pressures on the two sides of the eardrum before the Eustachian tube equalizes them. The eardrum cannot vibrate freely, causing the temporary symptoms described.

Question 4.4

Endoscopes are widely used in the diagnosis of colon diseases rather than other medical imaging methods because

- (1) it can provide a direct picture with fairly good resolution and clarity.
- (2) tools can be inserted through the duct in the endoscope to obtain tissue for further tests.
- (3) its risk is lower than that of other imaging methods.

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

Official answer: C

Endoscopy directly views the inner surface and permits tissue sampling through an instrument channel. It is invasive, so one cannot generally claim lower risk than all other imaging methods. Thus (1) and (2) are the valid reasons.

Question 4.5

A student measures the sound intensity level in dB at a distance of x from a small sound source. Estimate how far the student should be from the sound source in order to reduce the measured sound intensity level by 20 dB.

- A. $5x$
- B. $10x$
- C. $20x$
- D. $40x$

Official answer: B

$\Delta L = 10 \log(I_2/I_1) = -20 \text{ dB}$ gives $I_2/I_1 = 10^{-2}$. For a small source with inverse-square spreading, $I \propto 1/r^2$, so $r_2/r_1 = \sqrt{100} = 10$. The required distance is $10x$.

Question 4.6

For scanning the liver which is located inside the body, which of the following choices of ultrasound, with a reason, is correct?

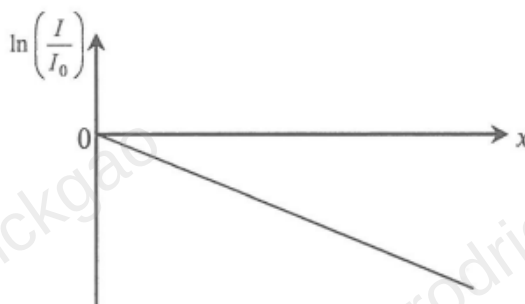
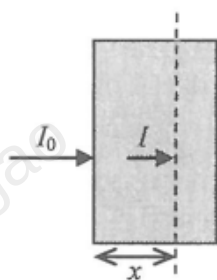
- A. 3 MHz ultrasound, as the image is of a higher resolution.
- B. 3 MHz ultrasound, as it can travel deeper inside the body.
- C. 12 MHz ultrasound, as the image is of a higher resolution.
- D. 12 MHz ultrasound, as it can travel deeper inside the body.

Official answer: B

Lower-frequency ultrasound is attenuated less strongly and can penetrate deeper. Thus 3 MHz is the appropriate choice for the deep liver, with penetration as the reason. Higher frequency improves resolution but reduces penetration depth.

Question 4.7

An X-ray beam of intensity I_0 is incident on a medium of linear attenuation coefficient μ . After travelling a distance x in that medium as shown, the intensity of the beam becomes I . A graph of $\ln(I/I_0)$ is plotted against x . What does the magnitude of the slope of the graph represent?



- A. $\mu/2$
- B. $(\ln 2)/\mu$
- C. $1/\mu$
- D. μ

Official answer: D

Attenuation follows $I = I_0 e^{-\mu x}$. Taking natural logarithms gives $\ln(I/I_0) = -\mu x$. The slope is $-\mu$, so its magnitude is μ , not the half-value thickness $(\ln 2)/\mu$.

Question 4.8

Which of the following statements about computed tomography (CT) scan is/are correct?

- (1) The grey levels in CT images correspond to X-ray attenuation coefficient of the body tissue.
 - (2) CT image reconstruction involves back projecting the intensity readings of the X-ray beam across an image plane viewed at different angles.
 - (3) The radiation dose received by a patient taking a CT scan is much higher than that received in conventional X-ray imaging.
- A. (1) only
 - B. (1) and (2) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

Official answer: D

CT reconstructs a map of X-ray attenuation from measurements at many angles; grey levels represent the reconstructed attenuation. The multiple projections generally give a substantially larger dose than a single conventional radiograph. All three statements are accepted in this examination.

Question 4 Structured Question

(a) Images A, B and C below were obtained by different medical imaging methods.



A (kidney)



B (body)

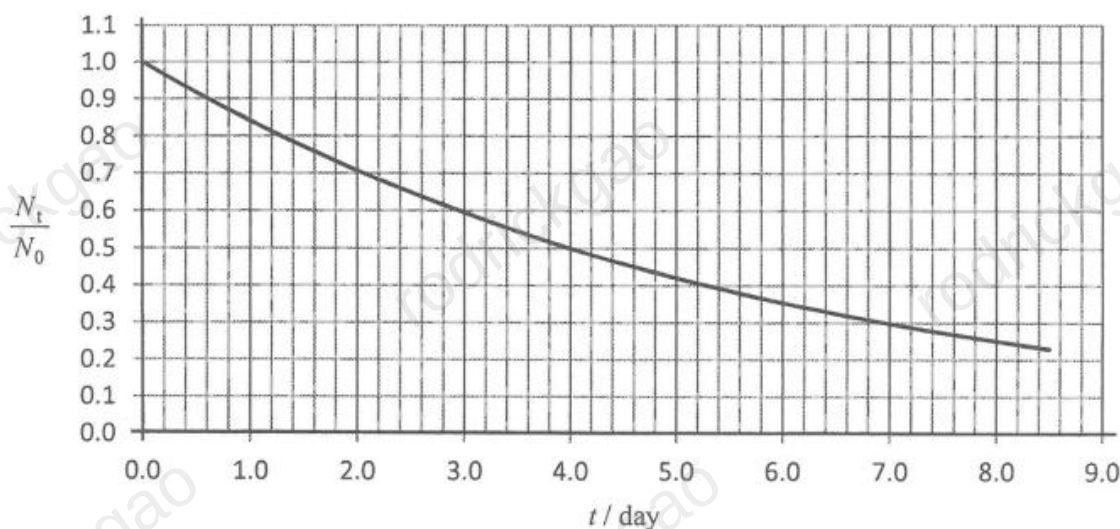


C (chest)

(a)(i) Which one is produced by radionuclide imaging? Explain how this image is formed. No need to describe the structure and mechanism of the detecting instrument used. (4 marks)

(a)(ii) State ONE advantage of radionuclide imaging over the other two imaging methods. (1 mark)

(b) A radioactive isotope of initial amount N_0 decays to become N_t after time t . The graph below shows the variation of the ratio N_t/N_0 with time t .



(b)(i) Use the graph to find the half-life of the radioactive isotope. (1 mark)

A chemical compound containing this radioactive isotope is used as a 'tracer' for injecting into a patient to study a physiological process. The biological half-life of this 'tracer' is 2 days.

(b)(ii) What is meant by the biological half-life of the 'tracer'? (1 mark)

(b)(iii) If 50 mg of this 'tracer' is injected initially, estimate the time taken for the amount of this radioactive chemical compound remaining in the body to drop to 10 mg. (3 marks)

Official marking scheme

Solution	Marks	Remarks
4. (a) (i) <i>B</i> A radioactive / radiopharmaceutical substance is injected to / inhaled by the patient and is transported in the blood stream to the rest of the body, the (radioactive) substance accumulates in particular organs. Gamma rays emitted by the radioisotope are detected by gamma cameras.	1A 1A 1A 1A 4	
(ii) Advantage: A hot spot (above normal uptake) or a cold spot (below normal uptake) can infer some problem with the organ, i.e. functional diagnosis.	1A 1	
(b) (i) $T_{\text{phy}} = 4$ days (ii) Period of time required to reduce the amount of 'tracer' in the body / organ to one-half of its original value due to biological process (such as elimination / excretion / metabolism).	1A 1 1A 1	
(iii) $\frac{1}{T_{\text{eff}}} = \frac{1}{T_{\text{phy}}} + \frac{1}{T_{\text{bio}}}$ $= \frac{1}{2} + \frac{1}{4}$ $T_{\text{eff}} = 1.33 \text{ days}$ $N = N_0 e^{-kt} \quad \text{or} \quad \left. \begin{aligned} \frac{N}{N_0} &= \left(\frac{1}{2}\right)^n \\ t &= nT_{\text{eff}} \end{aligned} \right\}$ $k = \frac{\ln 2}{T_{\text{eff}}}$ $t = \frac{-T_{\text{eff}} \ln(\frac{10}{50})}{\ln 2} \text{ or } 10 = 50 \left(\frac{1}{2}\right)^{t/T_{\text{eff}}}$ $t = \frac{-1.33 \ln(\frac{10}{50})}{\ln 2} \text{ or } 10 = 50 \left(\frac{1}{2}\right)^{t/1.33}$ $\approx 3.096 \text{ days}$	1M 1M 1A 3	Or 74.3 hours, $2.67 \times 10^5 \text{ s}$

(a)(i) Image B. A radioactive tracer is injected into or inhaled by the patient and transported by the bloodstream. It accumulates in particular organs according to physiological activity. Gamma rays emitted by the tracer are detected by a gamma camera to map its distribution.

(a)(ii) It reveals organ function through tracer uptake, including abnormally high or low uptake, rather than only showing anatomical structure.

(b)(i) At $N/N_0 = 0.5$ the graph gives $t = 4$ days, the physical half-life.

(b)(ii) Biological half-life is the time for biological processes alone, such as excretion or metabolism, to remove half the tracer initially present in the body or organ.

(b)(iii) Radioactive decay and biological removal act together. $1/T_{\text{eff}} = 1/T_{\text{phy}} + 1/T_{\text{bio}} = 1/4 + 1/2$, so $T_{\text{eff}} = 4/3$ days.

Using exponential decay, $t = T_{\text{eff}} \ln(50/10)/\ln 2 = (4/3) \ln 5/\ln 2 = 3.096 \text{ days} \approx 3.10 \text{ days}$. The result must be shorter than for physical decay alone.